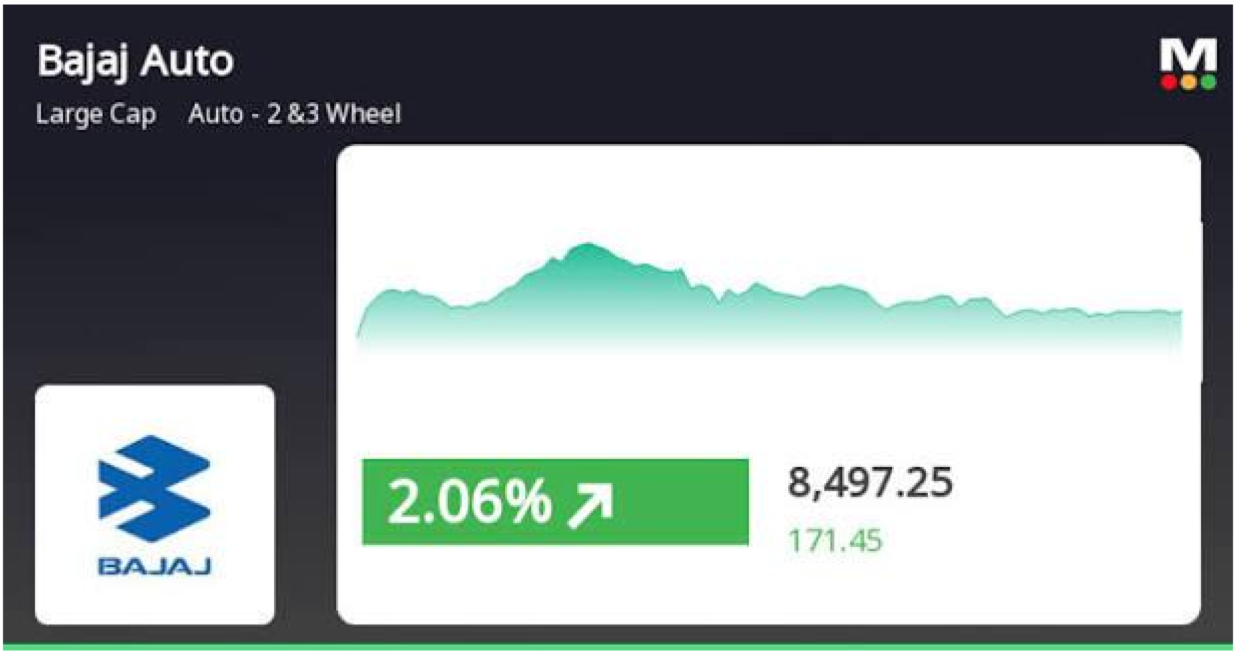
ANALYSIS OF BAJAJ AUTO PRICE & VOLUME MOVEMENT

This project presents a comprehensive visual analysis of Bajaj Auto using Tableau. The study integrates stock market behaviour, trading patterns, investor sentiment, and operational fundamentals through a 10-sheet interactive storyline. The analysis begins with price and volatility trends, moves into volume dynamics and investor behaviour, and concludes with geographic insights into Bajaj Auto's manufacturing footprint and plant productivity. The dashboards collectively provide a 360-degree understanding of how market performance aligns with the company's operational strength.

Introduction:

Bajaj Auto is one of India's leading two-wheeler manufacturers and a key player in the automobile industry. Its stock reflects not just market dynamics but also the company's production capacity, investor sentiment, and external economic conditions. This project aims to analyze Bajaj Auto's stock performance using Tableau, transforming raw daily stock data into meaningful visual insight.



The storyline moves from long-term price trends to daily fluctuations, volatility behaviour, trading volumes, quality of trades, and finally to business fundamentals through plant location and productivity mapping.

The purpose of this analysis is to offer a structured, data-driven perspective that connects stock market behaviour with real operational factors.

Objectives:

- To analyze Bajaj Auto's stock price trend and understand long-term market direction.
- To study daily price behaviour and identify periods of volatility and stability.
- To examine annual and monthly trading activity to identify participation patterns.
- To differentiate between speculative trading and genuine investor delivery.
- To detect high-impact outlier days in terms of price and volume behaviour.
- To analyze the seasonality of deliverable ratios to understand investor sentiment cycles.
- To visualize Bajaj Auto's manufacturing footprint using geospatial mapping.
- To assess relative productivity across Bajaj Auto's key manufacturing plants.
- To build a cohesive, sequential Tableau storyline that integrates market data with business fundamentals.

Business Questions:

Q1: How has Bajaj Auto's stock price evolved over time, and what long-term trends can be identified?

Q2: What does the daily price movement reveal about short-term market sentiment and trading behaviour?

Q3: During which periods did Bajaj Auto experience high or low volatility, and what factors may have contributed?

Q4: How has annual trading activity changed over the years, and what does this indicate about investor interest?

Q5: Are there specific months when Bajaj Auto consistently sees higher or lower trading activity?

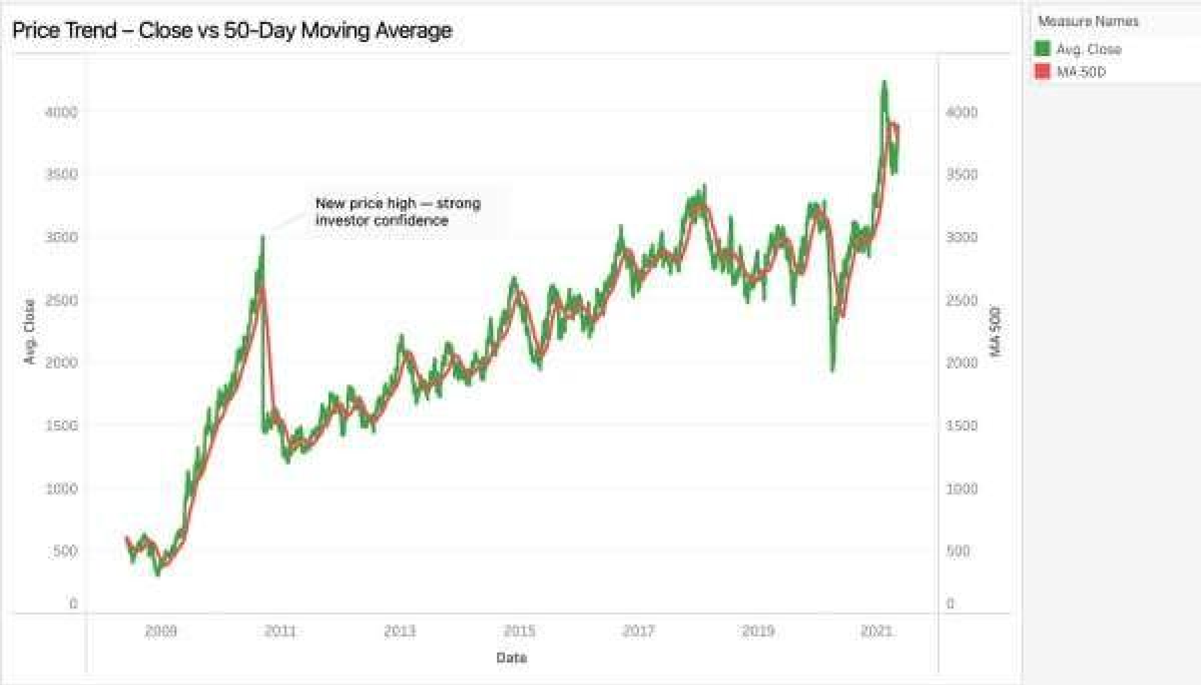
Q6: What proportion of trading activity results in actual delivery, and how does this reflect genuine investor participation?

Q7: Which trading days show abnormal behaviour in terms of volume and returns, and what could these outliers signify?

Q8: How does the deliverable percentage vary across months, and what does it reveal about long-term vs speculative trading?

Q9: How does productivity differ across Bajaj Auto's plants, and which units contribute the most to total output?

Q1: How has Bajaj Auto's stock price evolved over time, and what long-term trends can be identified?



Key Insights

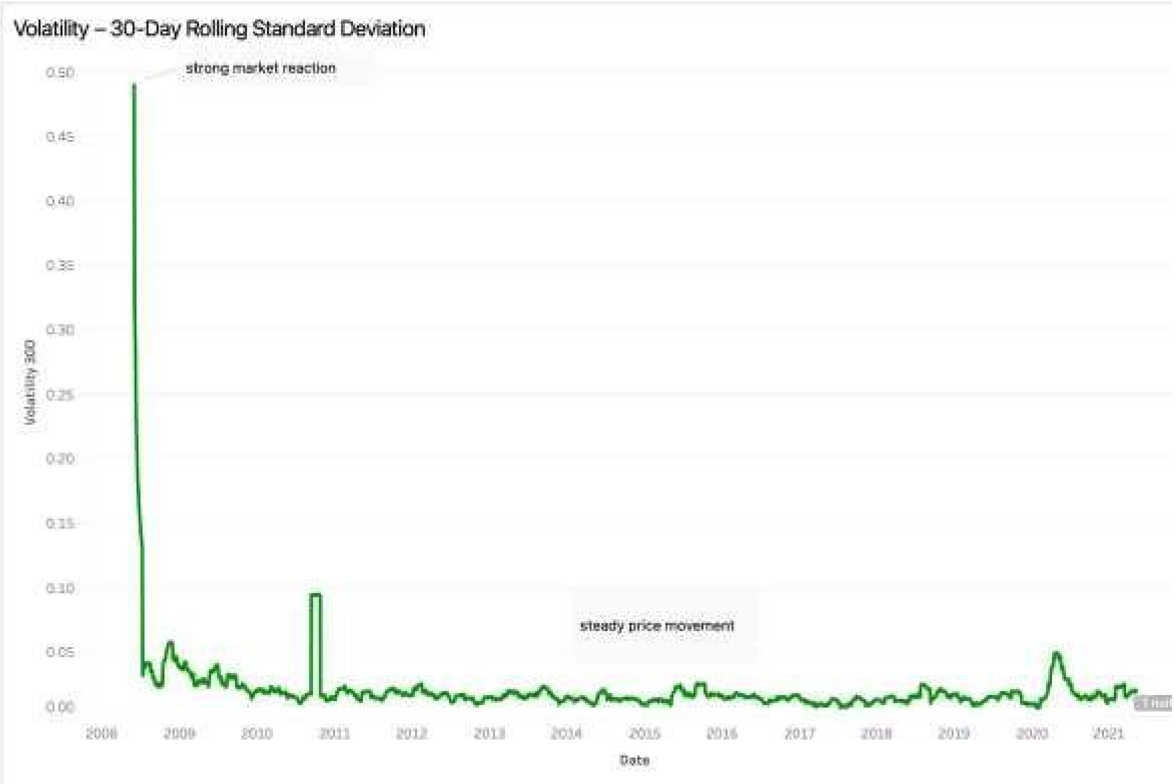
1. **Long-term Uptrend (2008–2021)**
   The stock shows a strong upward trajectory over the years, indicating long-term value creation and consistent investor confidence.
2. **MA50 Tracks the Trend Smoothly**
   The 50-day moving average (red line) acts as a trend stabilizer—whenever the price moves sharply up or down, MA50 smooths it out and helps identify the underlying direction.
3. **Big Rally Around 2010–2011**
   A very sharp rise followed by a correction is visible. This represents a period of aggressive buying and a quick profit-booking phase.
4. **Higher Highs Formed Regularly**
   From 2013 to 2021, the stock repeatedly forms higher highs and higher lows—a classic sign of a healthy bullish trend.
5. **New Price High with Strong Confidence (Highlighted)**
   The labelled point shows the price breaking into a new high zone, suggesting strong bullish sentiment and increased investor participation.

Q2: What does the daily price movement reveal about short-term market sentiment and trading behaviour?

Key Insights

1. **Mixed Days but Bias Toward Positive Returns**
   While both up (green) and down (red) days occur regularly, the chart shows more frequent small positive returns—indicating steady growth.
2. **High Volatility Periods (2018 & 2020)**
   Large spikes—both positive and negative—appear during global/marketwide disruptions, reflecting emotional/fear-driven trading.
3. **Strong Positive Return (Marked)**
   A big green spike shows a day with exceptionally strong gains, likely driven by positive news, results, or market momentum.
4. **Largest Single-Day Drop (Marked)**
   The deepest red bar indicates the most significant crash day—likely during the COVID-19 shock (March 2020). This highlights extreme short-term risk.
5. **Average Line Shows Slight Positive Drift**
   The average return lies just above zero, meaning the stock generally moves upward over time despite periodic negative days.

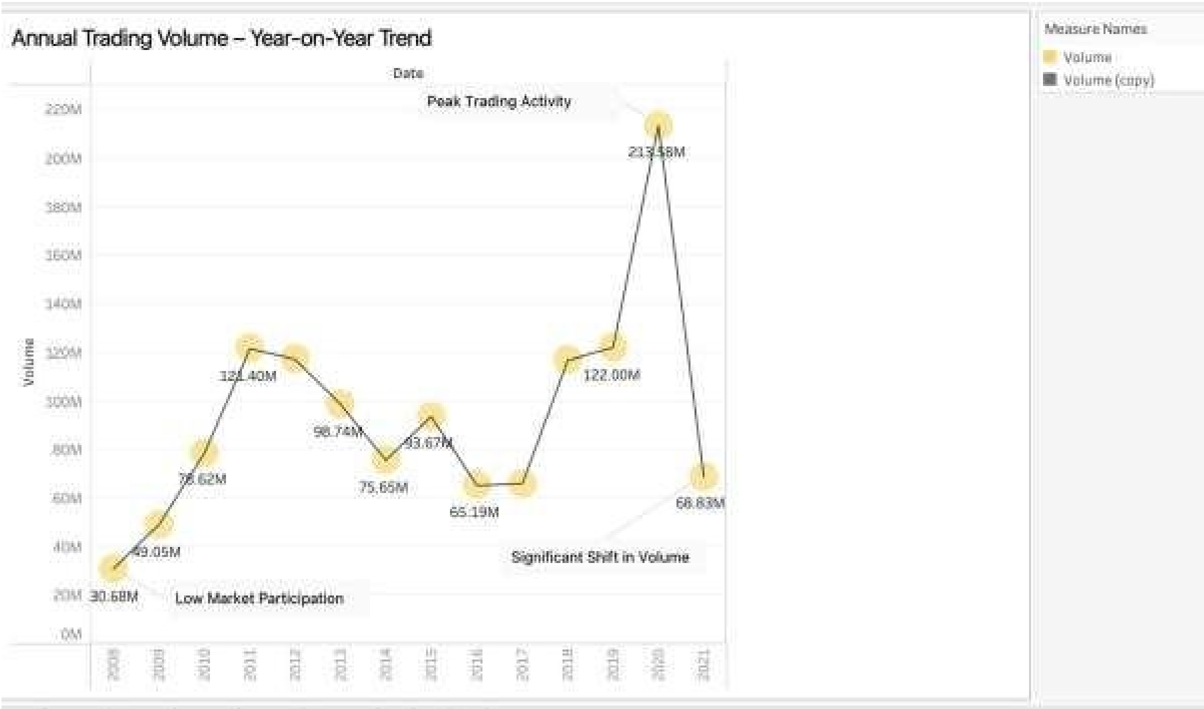
Q3: During which periods did Bajaj Auto experience high or low volatility, and what factors may have contributed?



Key Insights

1. **Extremely high volatility in 2008–09** shows a strong market reaction, likely driven by major global uncertainty and large price swings.
2. **A noticeable spike around 2011** indicates another short burst of instability or a sharp price event.
3. **From 2012 onward, volatility stays consistently low**, reflecting stable price movement and calmer market conditions.
4. **A moderate rise in 2020** aligns with COVID-19–related market disruptions, causing short-term uncertainty.
5. **Post-2020 volatility returns to normal**, showing the stock's strong recovery and regained market confidence.

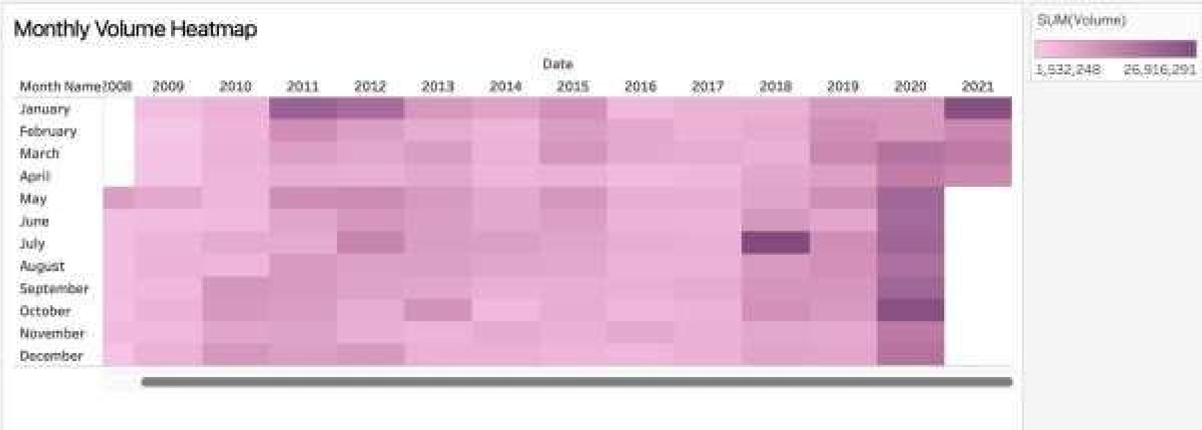
Q4: How has annual trading activity changed over the years, and what does this indicate about investor interest?



Key Insights

1. Low Market Participation in Early Years (2008–2010)
Trading volume starts relatively low, indicating limited investor activity and lower market interest during this period.
2. Strong Growth in Volume Around 2011–2012
Volume peaks around **121M** during 2011–12, showing increased investor participation and possibly strong news, performance, or market momentum.
3. Gradual Decline and Stabilization (2013–2017)
After the early peak, trading volume steadily declines and stabilizes between **65M–98M**, suggesting normalised activity with no major disruptions.
4. Significant Rise in 2018 and Extreme Spike in 2020
A dramatic volume surge occurs in **2020 (~213M)**—the highest in the entire period—indicating heightened market activity, likely driven by volatility and investor reactions during the COVID-19 period.
5. Drop in 2021, Returning to Normal Levels
Volume drops back to **68M in 2021**, showing normalization after the extraordinary 2020 spike and signalling reduced speculative activity.

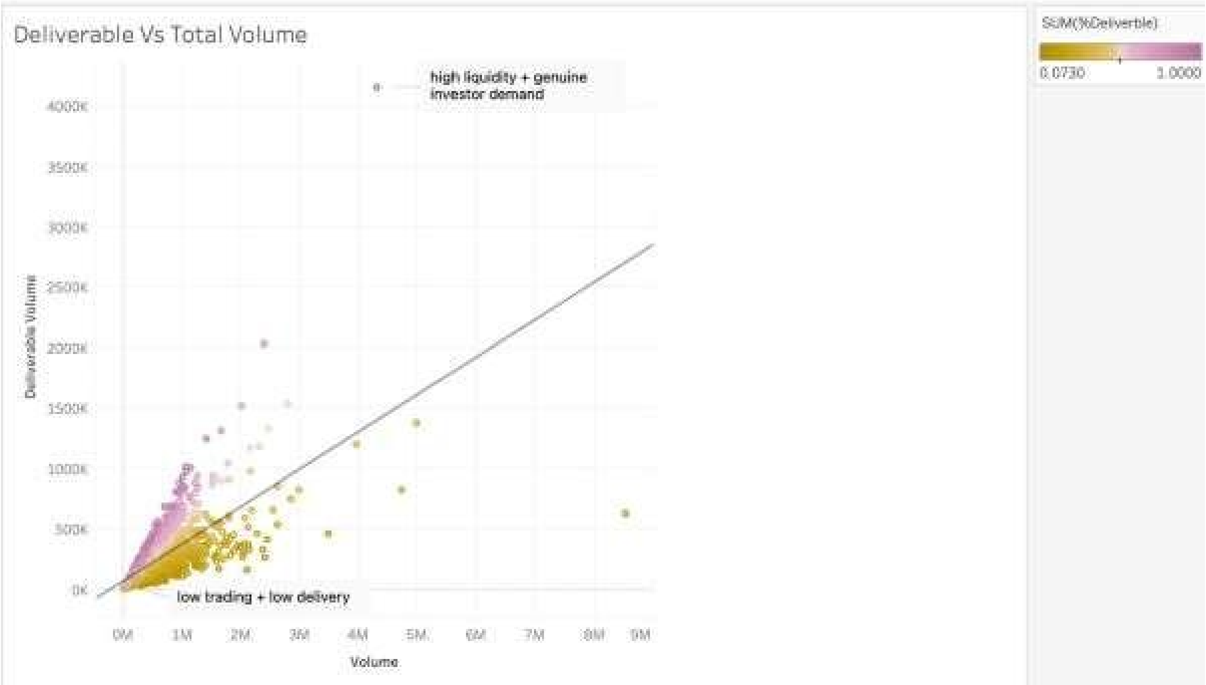
Q5: Are there specific months when Bajaj Auto consistently sees higher or lower trading activity?



Key Insights

1. **2011–2012 show the highest monthly activity**, especially in early months, indicating a period of strong investor participation.
2. **2018 and 2020 also show darker patches**, marking renewed spikes in trading interest.
3. **Monthly volumes from 2013–2017 remain mostly moderate**, reflecting stable but unspectacular trading activity.
4. **2020 shows consistent high activity across multiple months**, aligning with market-wide volatility during COVID-19.
5. **No strong seasonal pattern**, but occasional monthly hotspots suggest news-driven or event-driven trading.

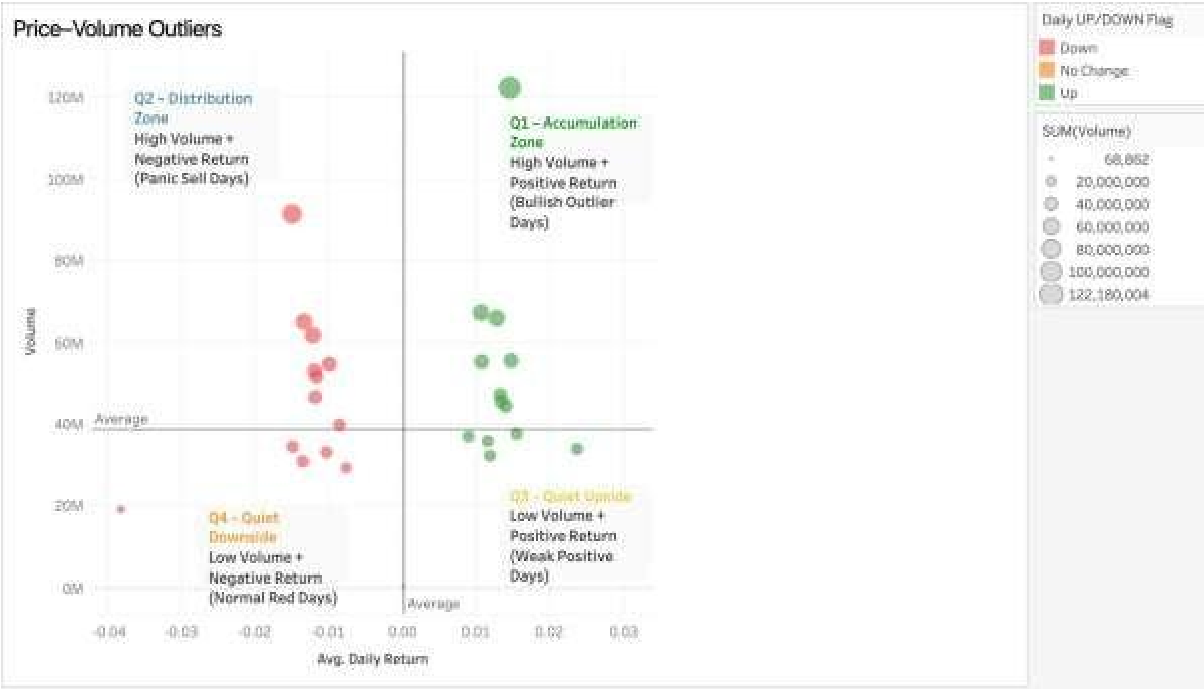
Q6: What proportion of trading activity results in actual delivery, and how does this reflect genuine investor participation?



Key Insights

1. **Most points lie near the lower-left**, showing many days with low trading volume and low delivery—typical for normal market activity.
2. **High deliverable % points (top-left cluster)** show strong investor confidence—buyers actually take shares into their demat accounts.
3. **High-volume but low-delivery points (bottom-right)** indicate speculative trading rather than long-term investment.
4. **Points above the trend line** represent **genuine demand**, where delivery rises faster than total volume.
5. **Points below the trend line** suggest **short-term participation**, with traders entering and exiting quickly.

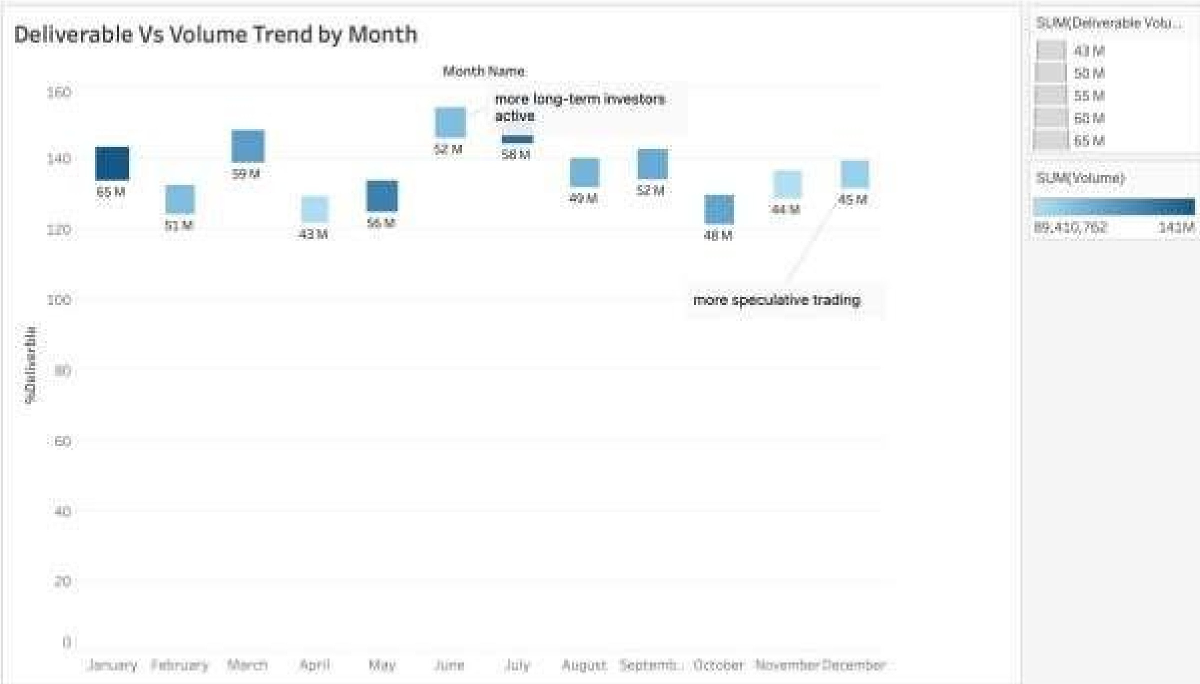
Q7: Which trading days show abnormal behaviour in terms of volume and returns, and what could these outliers signify?



Key Insights

1. **Q1 (High volume + positive return)** shows bullish outlier days where strong buying pushes the price up—**accumulation phase**.
2. **Q2 (High volume + negative return)** indicates panic selling or distribution—**strong selling pressure** on large volumes.
3. **Q3 (Low volume + positive return)** reflects quiet price gains, typically on low participation—**weak but positive sentiment**.
4. **Q4 (Low volume + negative return)** captures normal red days with limited trading—**mild downside, not panic**.
5. The chart shows **clear behaviour clusters**, helping identify market phases: accumulation, distribution, quiet upside, and quiet downside.

Q8: How does the deliverable percentage vary across months, and what does it reveal about long-term vs speculative trading?



Key Insights

1. **Higher deliverable % in months like January & March** indicates stronger long-term investor participation during these periods.
2. **Lower deliverable % in months like September–October** suggests more short-term/speculative trading activity.
3. **Overall deliverables remain above 120%**, showing consistent interest from genuine buyers rather than pure traders.
4. **Monthly fluctuations hint at sentiment shifts**, where certain months attract more committed investors.
5. The trend helps identify **ideal periods for corporate communication or major announcements**, aligning with stronger investor engagement.

Q9: How does productivity differ across Bajaj Auto's plants, and which units contribute the most to total output?

Key Insights

1. **Three major plants—Chakan, Waluj, and Pantnagar—cover strategic regions of India**, ensuring efficient nationwide distribution.
2. **Chakan Plant has the highest capacity (3.2M units)**, making it the central production hub for high-demand models.
3. **Pantnagar Plant (2.8M units)** serves as the key supplier for northern markets, reducing logistics time and transportation cost.
4. **Waluj Plant (1.5M units)** adds manufacturing flexibility and supports operational continuity across the network.
5. **Well-distributed plant capacity reduces risk**, strengthens supply chain resilience, and supports both domestic and export demand effectively.